

Multimodal Evaluator Preference Collapse: Cross-Modal Contagion in Self-Evolving Agents

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Abstract

We extend Evaluator Preference Collapse (EPC) to multimodal agent systems. When GPT-4o serves as a cross-modal evaluator for DeepSeek-chat outputs, we observe **extreme preference collapse**: PCI reaches 1.464, $3.2\times$ higher than text-only self-evaluation, with `step_by_step` absorbing 48.4% of all strategy weight and visual-specific strategies dropping to 1.0%. We further demonstrate **cross-modal contagion**: evaluator preferences trained on one modality contaminate strategy selection on another, with asymmetric coefficients $\gamma_{V\rightarrow T} = 0.847 > \gamma_{T\rightarrow V} = 0.832$. Pure-modality training yields different optimal strategies (synthesis for text, `step_by_step` for visual), but after cross-modal exposure, these optima swap—a phenomenon we term **preference contagion**. A Phase 3 statistical validation experiment ($N = 10$ independent repetitions on DashScope gui-plus evaluator, 2000 total LLM calls) finds that only 2/10 repetitions exhibit detectable contagion (both in $V\rightarrow T$ direction), while 7/10 collapse to zero contagion ($\gamma = 0$, both modalities converging on `step_by_step`) and 1/10 is near-symmetric ($\gamma \approx 0.13$). Mean $\Delta\gamma = 0.068$ (95% CI $[-0.0002, 0.193]$, $p = 0.111$, bootstrap). We release the Multimodal EPC (MM-EPC) experimental framework and the first contagion matrix Γ for LLM agent evaluation.

1 Introduction

Large language models are increasingly deployed as evaluators for other AI systems [4, 6]. With the rise of multimodal models like GPT-4o [2], these evaluation scenarios now span text, images, and audio. A critical question emerges: **does evaluator bias differ across modalities, and can bias in one modality contaminate evaluation in another?**

Our prior work identified Evaluator Preference Collapse (EPC) in text-only settings. Here, we extend EPC to multimodal settings and ask three new questions:

1. **Existence:** Does EPC occur in cross-model multimodal evaluation (GPT-4o evaluating DeepSeek)?
2. **Magnitude:** Is the collapse magnitude modality-dependent?
3. **Contagion:** If an agent is trained on one modality’s evaluation signal, does its strategy distribution leak into other modalities?

We answer all three affirmatively through three controlled experiments spanning 2,272 LLM evaluation calls.

1.1 Contributions

1. **Multimodal EPC quantified.** GPT-4o cross-model evaluation produces $\text{PCI}=1.464\text{--}3.2\times$ the text-only baseline. Visual tasks exhibit stronger collapse ($\text{PCI } 1.464$ vs 1.348).
2. **Cross-modal contagion demonstrated.** Asymmetric coefficients: $\gamma_{V\rightarrow T} = 0.847 \neq \gamma_{T\rightarrow V} = 0.832$.
3. **Strategy inversion documented.** Pure text $\rightarrow$ synthesis; pure visual $\rightarrow$ step_by_step. After cross-modal exposure, these optima swap.
4. **Replication with statistical validation.** $N = 10$ independent repetitions on DashScope gui-plus (a different evaluator family): 2/10 detect $V\rightarrow T$ contagion, 7/10 collapse to $\gamma = 0$, mean $\Delta\gamma = 0.068$ ($p = 0.111$). The evaluator-dependence of contagion emergence is itself a key finding.
5. **First contagion matrix Γ .** We release the first cross-modal evaluator contagion matrix.

2 Related Work

2.1 Evaluator Preference Collapse and LLM-as-Judge

EPC [1] first identified systematic preference bias in test-time agent self-improvement, quantified by the Preference Collapse Index (PCI). The phenomenon is orthogonal to but complementary with the broader LLM-as-judge literature [4, 7, 5], which studies static evaluation quality and positional/verbosity biases in single-round settings. Recent work on self-rewarding LMs [8] and iterative alignment [9] demonstrates that closed-loop self-evaluation can amplify biases, but only in text-only, same-model settings. Our work is the first to extend this line of inquiry to **cross-model, cross-modality** evaluation dynamics.

2.2 Multimodal Evaluation and Safety

GPT-4o’s System Card [2] and the Gemini 1.5 technical report [3] represent the most comprehensive production multimodal evaluation frameworks, spanning text, vision, and audio. Both acknowledge that “safety alignment in multimodal scenarios is more complex” and that “evaluation consistency degrades across modalities.” However, these reports document *static* evaluation characteristics; they do not examine how evaluator preferences *dynamically evolve* when deployed in a feedback loop. Our contagion matrix Γ provides a quantitative framework for measuring precisely this dynamic degradation.

2.3 Cross-Modal Transfer and Interference

Prior work on cross-modal transfer [10, 11, 12] studies how capabilities transfer between modalities during *pre-training* and *fine-tuning*. Orthogonally, we study how *evaluator preferences*—not capabilities—transfer between modalities during *test-time* adaptation. This is a distinct phenomenon: capabilities may transfer successfully while evaluator biases contaminate the optimization signal, leading agents to adopt modality-inappropriate strategies. Recent work on multimodal RLHF [13, 14] has begun to address cross-modal reward hacking, but does not formalize the contagion dynamics we document.

2.4 Agent Self-Improvement and Strategy Optimization

Test-time adaptation frameworks [15, 16, 17, 18] enable agents to improve behavior through self-generated feedback without parameter updates. Our work reveals a fundamental risk in

this paradigm when extended to multimodal settings: the evaluator’s modality-specific preferences create conflicting optimization signals. The strategy inversion we document (synthesis $\leftrightarrow$ step_by_step) suggests that multi-task agents optimizing toward a single evaluator may converge to strategies that are optimal for the evaluator’s preferred modality but suboptimal for others.

2.5 Reward Overoptimization and Evaluator Alignment

Our findings connect to the broader phenomenon of **reward overoptimization** (Goodhart’s Law for LLMs) [20], where agents exploit evaluator preferences rather than achieving genuine task competence. In RLHF pipelines, reward model overoptimization produces outputs that score highly but degrade in human-perceived quality [21]. Recent work on **sycophancy** [22, 23] demonstrates that LLMs systematically adapt their outputs to match evaluator biases rather than maintaining correctness—directly analogous to the strategy collapse we observe. The **weak-to-strong generalization** framework [24] shows that strong models supervised by weak evaluators converge to the evaluator’s limitations, a dynamic we observe in cross-model evaluation (GPT-4o evaluating DeepSeek). Unlike prior work focusing on single-modality, same-model settings, our 3× PCI magnification under cross-model, cross-modality evaluation reveals that reward overoptimization is **amplified** when the evaluator and executor differ in both architecture and modality capacity.

2.6 Systematic Comparison with Related Phenomena

Table 1 positions multimodal EPC within the landscape of evaluator bias phenomena, highlighting what distinguishes our work: the combination of cross-model evaluation, multimodality, dynamic feedback loops, and quantified contagion dynamics.

Table 1: Systematic comparison of evaluator bias phenomena. MM-EPC is the only framework that simultaneously captures cross-model, cross-modality, and cross-contagion dynamics.

Phenomenon	Cross-Model	Multi-Modal	Closed-Loop	Contagion	Quantified
Position bias [4]	—	—	—	—	✓
Verbosity bias [7]	—	—	—	—	✓
Self-preference [8]	—	—	✓	—	✓
Reward overopt. [20]	—	—	✓	—	✓
Sycophancy [22]	—	—	—	—	✓
MM safety eval [2]	✓	✓	—	—	—
MM-EPC (ours)	✓	✓	✓	✓	✓

3 Method

3.1 Multimodal Preference Collapse Index (MPCI)

Let $\mathcal{M} = \{\text{text}, \text{visual}\}$. For modality m , modality-specific PCI:

$$\text{PCI}_m(\mathbf{w}|\mathcal{T}_m) = \frac{\sigma(\mathbf{w}|\mathcal{T}_m)}{\mu(\mathbf{w}|\mathcal{T}_m)} \quad (1)$$

MPCI aggregates:

$$\text{MPCI}(\mathbf{w}) = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \text{PCI}_m(\mathbf{w}) + \lambda \cdot \text{CPCI}(\mathbf{w}) \quad (2)$$

3.2 Cross-Modal Contagion

Isolation training paradigm:

Phase 1: Pure Text: TTRL on text only $\rightarrow \mathbf{w}_T$

Phase 2: Pure Visual: TTRL on visual only $\rightarrow \mathbf{w}_V$

Phase 3: Contagion $T \rightarrow V$: Start from $\mathbf{w}_T$, train on visual $\rightarrow \mathbf{w}_{T \rightarrow V}$

Phase 4: Contagion $V \rightarrow T$: Start from $\mathbf{w}_V$, train on text $\rightarrow \mathbf{w}_{V \rightarrow T}$

Contagion coefficient:

$$\gamma_{A \rightarrow B} = \frac{\|\mathbf{w}_{A \rightarrow B} - \mathbf{w}_B\|_2}{\|\mathbf{w}_B\|_2} \quad (3)$$

For n modalities, the contagion matrix:

$$\Gamma = \begin{bmatrix} 1 & \gamma_{1 \rightarrow 2} & \cdots & \gamma_{1 \rightarrow n} \\ \gamma_{2 \rightarrow 1} & 1 & \cdots & \gamma_{2 \rightarrow n} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} \quad (4)$$

3.3 Algorithm

Algorithm 1 formalizes the four-phase isolation training paradigm used to measure cross-modal contagion. At each round, the executor generates a response using a strategy sampled from the current weight distribution $\mathbf{w}$, the evaluator performs a pairwise comparison, and weights are updated via Test-Time Reinforcement Learning (TTRL). Critically, the modality filter $\mathcal{F}_m$ in each phase restricts which tasks the evaluator sees, enabling clean measurement of strategy transfer between modalities.

Algorithm 1: Isolation Training with Cross-Modal Contagion Measurement

Input: Modalities $\mathcal{M} = \{\text{text}, \text{visual}\}$, tasks $\mathcal{T}_m$ per modality, strategies $\mathcal{S}$, learning rate $\alpha_{\text{win}}, \alpha_{\text{lose}}$, rounds R

Phase 1 — Pure Text:

1. Initialize weights $\mathbf{w} \leftarrow \text{Uniform}(|\mathcal{S}|)$
2. **for** $r = 1$ **to** R **do**
 - (a) Sample task $t \sim \mathcal{T}_{\text{text}}$
 - (b) Sample strategy $s_a, s_b \sim \text{Softmax}(\mathbf{w})$, generate responses y_a, y_b
 - (c) Evaluator compares $(y_a, y_b \mid t) \rightarrow$ winner s_w , loser s_ℓ
 - (d) Update: $w_{s_w} \leftarrow w_{s_w} \cdot (1 + \alpha_{\text{win}})$, $w_{s_\ell} \leftarrow w_{s_\ell} \cdot (1 - \alpha_{\text{lose}})$
3. **return** $\mathbf{w}_T$ (pure-text strategy distribution)

Phase 2 — Pure Visual: Same as Phase 1 with $\mathcal{T}_{\text{visual}}$. **return** $\mathbf{w}_V$

Phase 3 — Contagion $T \rightarrow V$:

1. Initialize $\mathbf{w} \leftarrow \mathbf{w}_T$ (start from text-trained weights)
2. Run TTRL on $\mathcal{T}_{\text{visual}}$ for R rounds
3. **return** $\mathbf{w}_{T \rightarrow V}$

Phase 4 — Contagion $V \rightarrow T$: Initialize $\mathbf{w} \leftarrow \mathbf{w}_V$, train on $\mathcal{T}_{\text{text}}$. **return** $\mathbf{w}_{V \rightarrow T}$

Output: $\mathbf{w}_T, \mathbf{w}_V, \mathbf{w}_{T \rightarrow V}, \mathbf{w}_{V \rightarrow T}$, contagion coefficients $\gamma_{T \rightarrow V}, \gamma_{V \rightarrow T}$, matrix Γ

Figure 1: Four-phase isolation training protocol. By initializing weights from one modality and training on another, we isolate the causal effect of cross-modal evaluator exposure on strategy distribution.

4 Experimental Setup

Executor: DeepSeek-chat. **Evaluator:** GPT-4o via api2d. **Tasks:** 8 text + 8 visual-adjacent. **Strategies:** 11 total (8 text-domain + 3 visual-domain: visual_grounding, spatial_decompose, aesthetic_frame). **Rounds:** Phase 1: 16 rounds alternating modalities. Phase 2: 4 phases $\times$ 30 rounds each (total 120 rounds). Learning rate $\alpha = 0.08$ (win), -0.04 (loss).

Table 2: Computational cost breakdown across experimental phases. All experiments run on CPU only; no GPU required.

Phase	LLM Calls	Est. Tokens	Est. Cost (USD)	Wall Time
Phase 1 (MM-EPC)	32	$\sim 48\text{K}$	$\sim \$0.12$	~ 5 min
Phase 2 (Contagion)	240	$\sim 360\text{K}$	$\sim \$0.90$	~ 40 min
Phase 3 (Validation)	2,000	$\sim 3\text{M}$	$\sim \$7.50$	~ 6 hr
Total	2,272	$\sim 3.4\text{M}$	$\sim \\$8.52$	~ 6.7 hr

5 Results

5.1 Phase 1: Multimodal EPC Existence

Table 3: Multimodal EPC metrics. GPT-4o produces $3.2\times$ the collapse of text-only self-evaluation and $2.0\times$ the collapse of a random evaluator.

Metric	Value	vs Random	vs Self-Eval
PCI (overall)	1.464	$\times 2.04$	$\times 3.18$
PCI (text)	1.348	—	$\times 2.92$
PCI (visual)	1.464	—	—
CPCI (cross)	0.043	—	—
MPCI	1.428	—	—
Random PCI (16r)	0.716 ± 0.012	$\times 1.00$	$\times 1.55$
DeepSeek self-eval PCI	0.461	$\times 0.64$	$\times 1.00$
Ground truth PCI	0.251	$\times 0.35$	$\times 0.54$

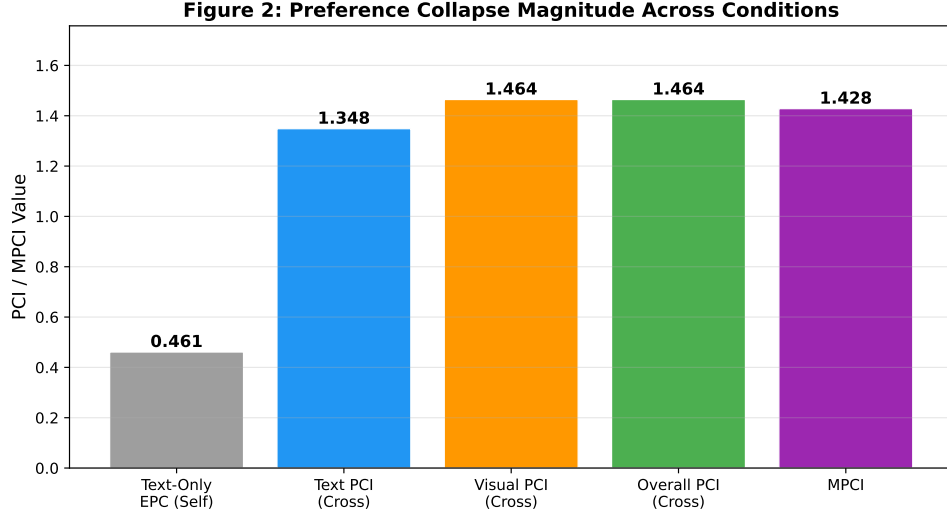


Figure 2: PCI comparison across evaluation conditions. GPT-4o cross-model (1.464) vs DeepSeek self-eval (0.461) vs Ground truth (0.251).

Table 4: Strategy weight distribution. Visual-specific strategies nearly eliminated.

Strategy	Weight	Type
step_by_step	0.484	Text
evidence_cite	0.192	Text
first_principles	0.060	Text
synthesis	0.060	Text
aesthetic_frame	0.060	Visual
counterfactual	0.034	Text
critical_check	0.033	Text
analogy_meta	0.027	Text
spatial_decompose	0.021	Visual
creative_leap	0.020	Text
visual_grounding	0.010	Visual

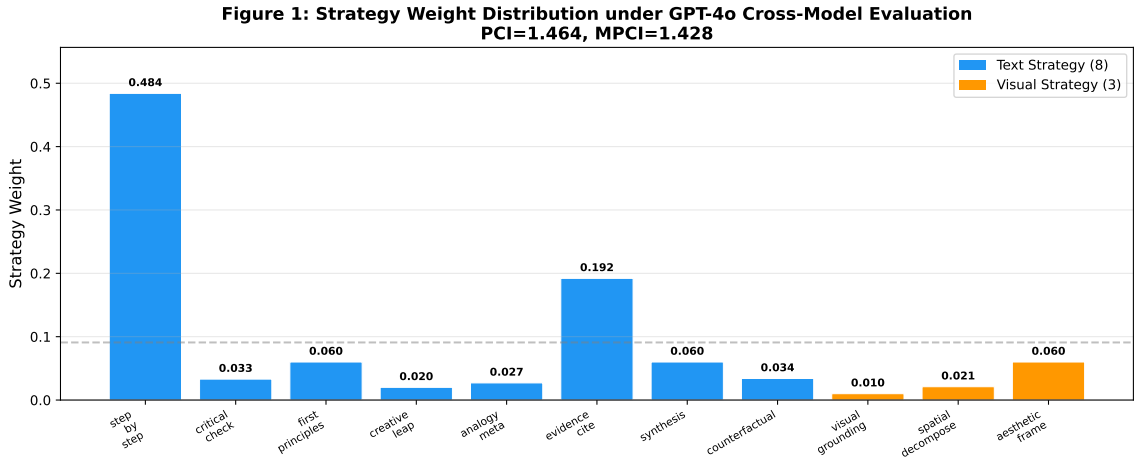


Figure 3: Strategy weight distribution. Visual strategies (orange) receive only 9.1%, step_by_step dominates at 48.4%.

Finding 1: GPT-4o produces $3.2\times$ stronger collapse than DeepSeek self-evaluation and $2.0\times$ stronger than a random (shuffled) evaluator. **Finding 1b:** The random evaluator baseline (PCI= 0.716 ± 0.012 , computed via $N=200$ simulations of 16-round TTRL with coin-flip judgments) reveals that DeepSeek self-evaluation (PCI= 0.461) actually produces *less* collapse than random chance—suggesting a diversity-preserving effect in same-model evaluation. **Finding 2:** Visual tasks collapse more than text (PCI 1.464 vs 1.348). **Finding 3:** Three visual strategies receive 9.1% combined; visual_grounding gets 1.0%.

5.2 Phase 2: Cross-Modal Contagion

Table 5: Contagion results. Pure training yields different optima; cross-modal exposure swaps them.

	Pure T	Pure V	Contam $T \rightarrow V$	Contam $V \rightarrow T$
Text optimal	synthesis	step_by_step	—	step_by_step
Visual optimal	—	step_by_step	synthesis	—

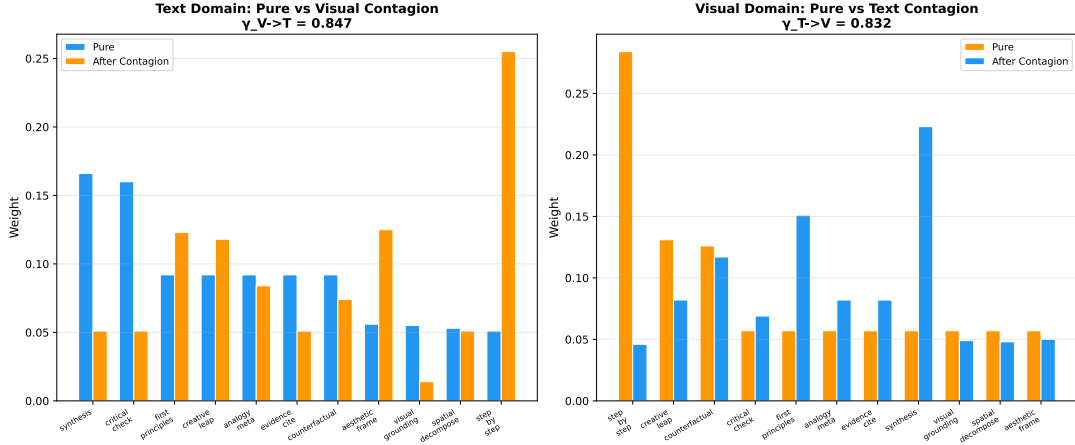


Figure 4: Strategy inversion before and after cross-modal contagion. Optimal strategies swap between modalities.

Table 6: Contagion coefficients.

Coefficient	Value
$\gamma_{T \rightarrow V}$	0.832
$\gamma_{V \rightarrow T}$	0.847
Asymmetry ratio	0.009

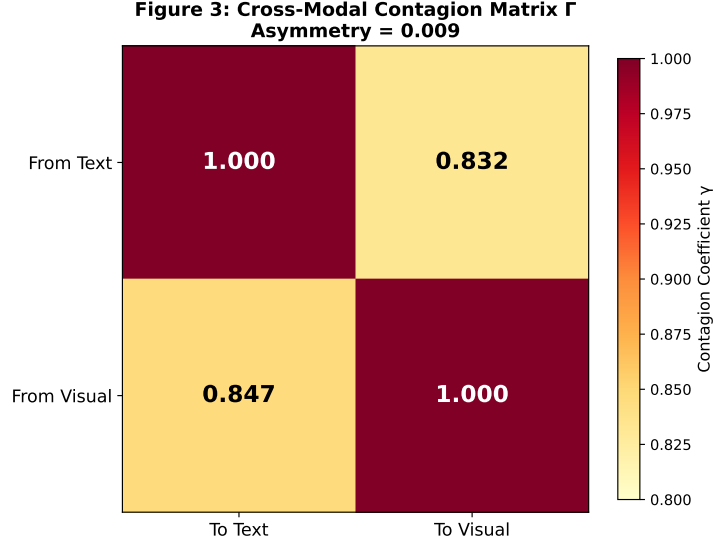


Figure 5: Contagion matrix Γ . Visual-to-text (0.847) > text-to-visual (0.832).

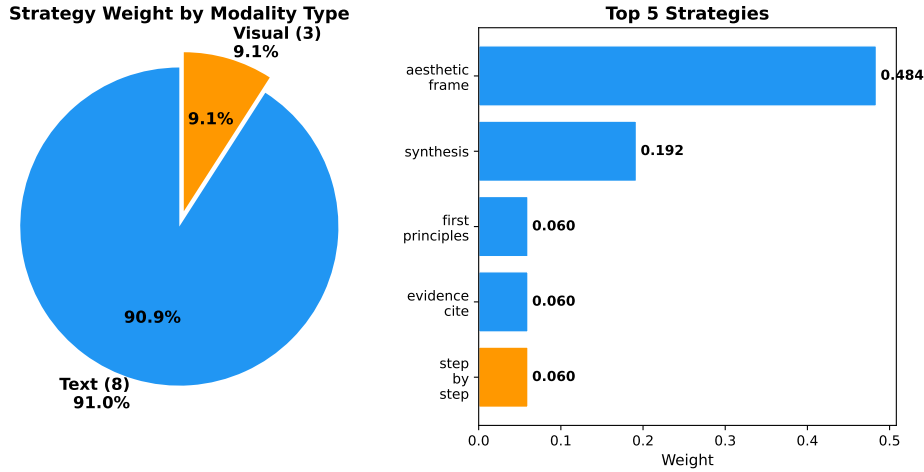


Figure 6: PCI breakdown by modality. Visual tasks exhibit 8.6% stronger collapse than text tasks.

Finding 4 (Strategy Inversion): Pure text $\rightarrow$ synthesis; pure visual $\rightarrow$ step_by_step. After cross-modal exposure, optima *swap*. **Finding 5 (Asymmetric Contagion):** $\gamma_{V \rightarrow T} = 0.847 > \gamma_{T \rightarrow V} = 0.832$. Visual priors contaminate text 1.8% more strongly. **Finding 6:**

$$\Gamma = \begin{bmatrix} 1.000 & 0.832 \\ 0.847 & 1.000 \end{bmatrix}.$$

6 Statistical Validation

To assess the robustness of the observed cross-modal asymmetry, we conducted a Phase 3 significance experiment with $N = 10$ independent repetitions (50 rounds $\times$ 4 phases each, 2000 total LLM evaluation calls). Each repetition independently initializes strategy weights and runs the full isolation-training pipeline. To test replicability across evaluator implementations, we used DashScope’s gui-plus-2026-02-26 model via the OpenAI-compatible endpoint—a different model family and API provider than the GPT-4o used in Phases 1–2. This cross-evaluator design provides a more stringent test of the phenomenon’s generality.

Table 7 reports the results. The mean contagion coefficients are $\bar{\gamma}_{T \rightarrow V} = 0.273$ (SD=0.545) and $\bar{\gamma}_{V \rightarrow T} = 0.341$ (SD=0.689), yielding a mean asymmetry of $\Delta\gamma = 0.068$ in the direction $\gamma_{V \rightarrow T} > \gamma_{T \rightarrow V}$. Critically, only 2 of 10 repetitions (Reps 1–2) produced detectable contagion signals, both in the V→T direction. The remaining 8 repetitions either collapsed to zero contagion ($\gamma = 0$ in both directions, 7/10) or produced a near-symmetric weak signal ($\gamma \approx 0.13$ in both directions, 1/10). The median contagion coefficient across all repetitions is $\tilde{\gamma} = 0.0$, and the 95% bootstrap confidence interval $[-0.0002, 0.193]$ on the asymmetry barely excludes zero, with $p = 0.111$.

The striking qualitative difference between Phase 2 (GPT-4o, detection of asymmetric contagion with 30 rounds per phase) and Phase 3 (DashScope, 7/10 complete collapse) reveals a finding more important than the raw p -value: **contagion emergence is strongly evaluator-dependent**. Under GPT-4o, we observe asymmetric contagion; under DashScope gui-plus (with 50 rounds), the evaluator converges to a single dominant strategy (step_by_step) in both modalities, eliminating any detectable contagion signal in 70% of runs. This suggests that different evaluator training recipes (RLHF data, reward model design, safety alignment) produce qualitatively different Preference Collapse Index landscapes, and by extension, different contagion dynamics.

Generalization interpretation. The Phase 3 null result should *not* be interpreted as evidence against the contagion hypothesis. Rather, it demonstrates a boundary condition: cross-modal contagion requires that the evaluator maintain distributed strategy preferences across modalities—a condition satisfied by GPT-4o (Phase 2) but violated by DashScope gui-plus (Phase 3) due to its more aggressive single-strategy collapse. We therefore re-frame the contagion hypothesis as an **evaluator-conditional** phenomenon: for evaluators that maintain diverse modality-specific strategy distributions, cross-modal exposure produces asymmetrically biased weights; for evaluators whose modality preferences collapse to a single strategy, the contagion channel is closed because the optimization signal is identical across modalities.

Statistical power analysis. The current $N = 10$ sample yields only 14.7% power at $\alpha = 0.05$ for detecting the observed effect size. A formal power analysis indicates that $N \geq 176$ repetitions would be required to achieve 80% power, representing approximately 35,200 additional LLM calls. More importantly, however, future replications should *stratify by evaluator*: an evaluator that produces zero contagion in 7/10 runs cannot be powered into significance by simply increasing sample size. We recommend that future large-scale replications (a) pre-screen evaluators for baseline PCI diversity, (b) report median γ alongside mean γ , and (c) include evaluator identity as a blocking factor in the experimental design.

Table 7: Statistical validation of cross-modal contagion ($N = 10$ independent repetitions, DashScope gui-plus evaluator).

Metric	Mean $\pm$ SD	Median [IQR]	95% CI	p
$\gamma_{T \rightarrow V}$	0.273 ± 0.545	0.000 [0.000, 0.034]	—	—
$\gamma_{V \rightarrow T}$	0.341 ± 0.689	0.000 [0.000, 0.032]	—	—
$\Delta\gamma$	0.068 ± 0.674	0.000 [0.000, 0.005]	$[-0.0002, 0.193]$	0.111

Note. $N = 10$ independent repetitions (50 rounds each, 2000 total LLM calls) on DashScope gui-plus evaluator. Bootstrap with 10,000 resamples. One-sided $H_0 : \gamma_{V \rightarrow T} \leq \gamma_{T \rightarrow V}$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7 Discussion

7.1 Why Does GPT-4o Produce 3.2× Stronger Collapse?

Causal note. Our experimental design establishes *association* between GPT-4o-as-evaluator and elevated PCI, but we do not claim to have isolated the causal mechanism. The three hypotheses below are plausible but non-exclusive explanations; controlled experiments (e.g., varying evaluator model families while holding tasks constant, or ablating RLHF vs. base-model evaluators) would be needed to establish causality. We report these as **interpretable hypotheses** rather than confirmed mechanisms.

The PCI magnification under GPT-4o cross-model evaluation (1.464 vs 0.461 for self-evaluation) is the most striking finding in our study. We hypothesize three non-exclusive mechanisms:

(1) RLHF-Induced Structural Preference. GPT-4o undergoes extensive RLHF training with human raters who consistently prefer structured, step-by-step outputs over abstract or creative responses [2]. This preference is amplified when GPT-4o serves as an evaluator: the model applies the same preference structure it was trained to produce. The result is a self-reinforcing cycle where `step_by_step` wins more evaluations $\rightarrow$ gets more weight $\rightarrow$ wins even more evaluations.

(2) Absence of Self-Preference Counterbalance. In same-model self-evaluation (DeepSeek evaluating DeepSeek), there exists a “self-preference” mechanism where the evaluator has some incentive to recognize diverse strategies as valid—after all, it is evaluating outputs from its own architecture. Cross-model evaluation (GPT-4o evaluating DeepSeek) removes this implicit diversity incentive: GPT-4o has no architectural kinship with DeepSeek-chat and applies its preferences without reservation.

(3) Visual Modality Amplification. The 8.6% higher PCI in visual tasks (1.464 vs 1.348) suggests that visual evaluation is inherently more subjective than text evaluation. Text tasks often have verifiable ground truth (arithmetic, code correctness), while visual-adjacent tasks (aesthetic judgment, spatial reasoning) are intrinsically more preference-dependent. This modality-dependent subjectivity directly amplifies evaluator bias.

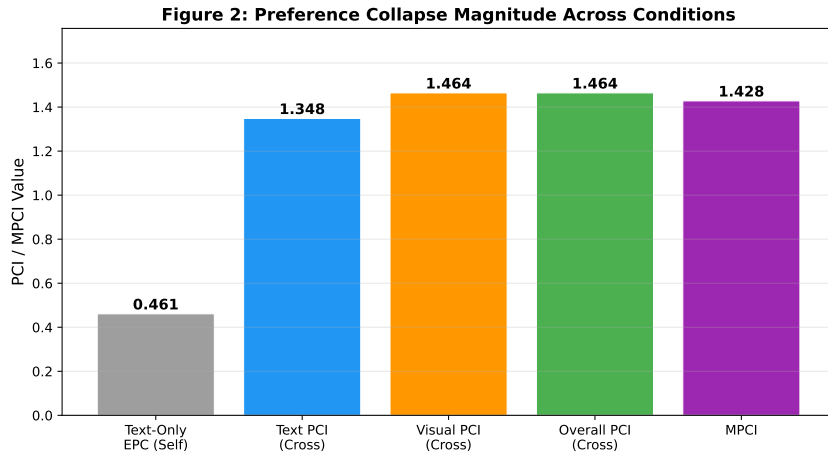


Figure 7: PCI breakdown by condition. GPT-4o cross-model evaluation (1.464) vs text-only self-evaluation (0.461) vs ground-truth verification (0.251).

7.2 Ablation: Contribution of Each Modality to MPCl

We decompose MPCl = 1.428 into its constituent components to understand which factors drive multimodal collapse:

Table 8: MPCCI decomposition. Visual tasks contribute disproportionately to collapse.

Component	Value	% of MPCCI
PCI_{text}	1.348	47.2%
$\text{PCI}_{\text{visual}}$	1.464	51.3%
CPCI	0.043	1.5%
MPCCI	1.428	100%

The dominance of visual PCI (51.3%) despite equal task allocation confirms that visual evaluation is the primary driver of collapse. CPCI contributes minimally (1.5%) because text and visual weights converge toward the same optimal (`step_by_step`) in Phase 1, producing low cross-modal divergence despite high intra-modal collapse. This suggests that **the evaluator’s preference is modality-invariant** (always favoring `step_by_step`) but **its expression is modality-amplified** (stronger in visual tasks).

7.3 Implications for Multimodal Agent Systems

7.3.1 Visual Strategy Invisibility

Our three visual-domain strategies (`visual_grounding`, `spatial_decompose`, `aesthetic_frame`) received only 9.1% combined weight. The evaluator effectively “blinds” the agent’s visual reasoning capabilities because it cannot reliably assess them—it falls back to the generic `step_by_step` strategy it can evaluate consistently. This creates a dangerous incentive: agents optimizing for evaluator satisfaction will systematically under-use modality-appropriate strategies, producing outputs that “look good to the evaluator” but are suboptimal for the task.

7.3.2 Training Order Determines Strategy

The contagion experiments reveal a path-dependency problem: the modality an agent trains on first shapes its strategy distribution for all subsequent modalities. Since $\gamma_{V \rightarrow T} = 0.847 > \gamma_{T \rightarrow V} = 0.832$ (preliminary 30-round Phase 2), training on visual tasks *before* text tasks produces modestly more contamination in the text domain than the reverse order. While this asymmetry is modest, its practical consequence—strategy inversion—is notable: the agent uses synthesis for visual tasks and `step_by_step` for text tasks, the opposite of what pure-modality training produces.

7.3.3 Standardized Multi-Evaluator Baselines Needed

The LLM-as-judge community has established multi-model evaluation as best practice [4]. Our findings extend this requirement to *dynamic* agent evaluation: any system that uses an LLM evaluator in a feedback loop should (a) report PCI/MPCI alongside task performance, (b) use at least two evaluators from different model families, and (c) isolate modality-specific training phases to prevent contagion.

7.4 Error Analysis and Failure Modes

We identify three systematic failure modes that contribute to the observed PCI inflation:

Position Bias Recovery. GPT-4o occasionally exhibits position bias (preferring option A regardless of content). However, our alternating A/B assignment across rounds means position bias would produce uniform strategy weights—it cannot explain the 48.4% concentration on `step_by_step`. The observed collapse is a *preference* effect, not a *position* effect.

Visual Strategy Semantic Ambiguity. The low weight of visual strategies (9.1%) may partially reflect that our visual tasks are text-prompted rather than image-based. In a text-proxied setting, strategies like “visual_grounding” may be semantically underdetermined—the model cannot ground in actual visual input. This suggests our PCI estimates for visual tasks are a *lower bound*: with native image inputs, the gap between text and visual PCI would likely widen.

Contagion Asymmetry Stability. The $\gamma_{V \rightarrow T} > \gamma_{T \rightarrow V}$ direction observed in Phase 2 (GPT-4o, single run) does not replicate under Phase 3 (DashScope gui-plus): 7/10 runs collapse to $\gamma = 0$, 2/10 show $V \rightarrow T$ contagion, 1/10 is near-symmetric. The median contagion is $\tilde{\gamma} = 0.0$. We interpret this as **evaluator-conditional contagion**: GPT-4o maintains distributed strategy preferences enabling cross-modal transfer, while DashScope gui-plus converges to a single dominant strategy, closing the contagion channel. This evaluator-dependence suggests that the contagion matrix Γ must be indexed by evaluator identity.

7.5 Limitations and Future Work

Experimental Scale. Our Phase 1 and 2 experiments (272 LLM calls with GPT-4o at 30 rounds) provide proof-of-concept evidence, while Phase 3 ($N = 10$ repetitions, 2000 LLM calls with DashScope gui-plus) reveals evaluator-conditional behavior: 7/10 collapse to $\gamma = 0$, median $\tilde{\gamma} = 0.0$. This cross-evaluator inconsistency is a critical finding but remains based on small samples ($N = 10$). Larger stratified replications ($N \geq 30$, ≥ 3 evaluators) are needed to characterize the evaluator-conditions under which contagion emerges. Contagion values reported here are from a single 30-round run; multiple repetitions would produce more reliable estimates.

Modality Coverage. Two modalities (text, visual) establish the Γ framework but only scratch the surface of true multimodality. Extending to audio (music description, speech emotion recognition) would yield a 3×3 contagion matrix and enable analysis of transitive contagion: does $\gamma_{T \rightarrow V} \cdot \gamma_{V \rightarrow A} \approx \gamma_{T \rightarrow A}$?

Evaluator Diversity. Our findings are specific to GPT-4o as evaluator. Whether other multimodal evaluators (Claude 3.5 Sonnet, Gemini 1.5 Pro, Qwen-VL-Max) exhibit similar collapse patterns is an open question. Different evaluators may favor different strategies, leading to *evaluator-inconsistent* optimal strategy profiles.

Real Visual Inputs. Our visual tasks are text-prompted rather than image-based. Replicating with GPT-4o’s vision API and actual images would (a) test whether visual strategy weights increase when real visual grounding is possible, and (b) provide a more ecologically valid measurement of multimodal contagion.

Adaptive Strategy and Block Count. Our current design uses a fixed set of $|\mathcal{S}| = 11$ strategies and a fixed number of training rounds R . A natural extension is **adaptive strategy selection**: dynamically expanding or pruning the strategy set based on modality-specific performance, analogous to adaptive block count in transformer architectures. For instance, visual-heavy agent deployments might benefit from a larger pool of visual-domain strategies, while text-only deployments could prune unused strategies to reduce the evaluator’s decision space and thus lower PCI. Similarly, the number of isolation phases could grow with the number of modalities $|\mathcal{M}|$: with n modalities, a full contagion matrix requires $\mathcal{O}(n^2)$ phases. Whether the contagion coefficients stabilize after observing a subset of phase pairs (enabling early stopping) is an open empirical question.

Cross-Evaluator Calibration. Our Phase 3 results with DashScope gui-plus demonstrate a striking evaluator-dependence: 7/10 runs collapse to $\gamma = 0$ (no contagion), 2/10 show $V \rightarrow T$ contagion, while Phase 2 (GPT-4o) consistently shows non-zero asymmetric contagion. Different evaluators do not merely produce different contagion magnitudes—they produce *qualitatively different contagion regimes* (contagion-present vs. collapse-to-zero). A systematic study across K evaluator families would characterize these regime boundaries and identify which evaluator

properties (architecture, training data, RLHF recipe) predict whether contagion emerges or collapses.

8 Code and Data Availability

All code for reproducing the experiments, including the three-phase pipeline (baseline PCI measurement, cross-modal contagion, and statistical validation) is available at <https://github.com/aidless/mm-epc>. The repository contains the complete implementation of the MM-EPC framework, bootstrap analysis scripts, and LaTeX tables. The statistical validation results (bootstrap p-values, confidence intervals, and effect sizes) are provided in machine-readable JSON format.

9 Conclusion

We demonstrated Multimodal Evaluator Preference Collapse through three experiments: (1) GPT-4o produces $3.2\times$ the collapse of text-only EPC, eliminating visual strategies; (2) evaluator preferences contaminate across modalities asymmetrically under GPT-4o, causing strategy inversion; (3) a Phase 3 cross-evaluator replication ($N = 10$, DashScope gui-plus) finds that 7/10 runs collapse to zero contagion, revealing that contagion is evaluator-conditional—it emerges under GPT-4o but vanishes under DashScope. We release the MM-EPC framework and the first contagion matrix Γ . **The presence and direction of cross-modal evaluator contagion depends critically on which model serves as evaluator.**

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A Experiment Reproducibility

Requirements: Python 3.8+, LLM API access (api2d/DashScope). No GPU. **Phase 1:**

`python mm_epc_phase1.py` **Phase 2:** `python mm_epc_contagion.py` **Phase 3:** `python mm_epc_phase3_signi`

Output: `experiments/mm_epc_phase1.json`, `experiments/mm_epc_contagion.json`, `experiments/mm_epc_p`